Spring potential energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate stored spring energy and distinguish it from force.

SUCCESS CRITERIA - I can...

1. Use displacement in meters.
2. Apply the squared relationship for spring energy.
3. Explain how stored energy can become motion.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A pendulum loses 15 J gravitational energy while descending. Neglect losses. Find its kinetic-energy gain.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $U_s = \frac{1}{2}kx^2$; $|F_{spring}| = k|x|$.
s means spring. Compression and stretch both store energy relative to the natural length; energy and force are different quantities.

Worked reasoning: A spring with k=200 N/m stretched 0.10 m stores $\frac{1}{2}kx^2=1$ J. Doubling stretch to 0.20 m stores 4 J, not 2 J.

Your turn: Draw force-versus-stretch triangles and compare their areas.

Marzano check - circle: ① ② ③

03 Look and explain

The same spring is displaced x and 2 x. Compare stored energies and force magnitudes.



Mark equilibrium and displacement x.

The zigzag is the spring; the rectangle is the attached block.

Marzano check - circle: ① ② ③

04 Build the solution

Use k=100 N/m and x=.20 m. Find x^2, then $U=\frac{1}{2}\times100\times x^2$.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Use the idea

Why is energy positive for both positive and negative x even though spring force reverses?

Marzano check - circle: 1 2 3

06 Choose a method

A 4 N friction force opposes a 6 m displacement. Find work, including sign and units.

Marzano check - circle: 1 2 3

07 Exit check

A spring stores 3 J at a certain compression. How much does it store at double compression?

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Find the energy for stretches 0.05, 0.10, 0.15 m when k=200 N/m. Sketch the energy trend.

Marzano check - circle: 1 2 3